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Supermarket Sales Analysis



SQL DATA PROJECT



SHAHD'S LINKEDIN

DATA ANALYST

Executive Summary

Project Scope & Business Objectives

Project Goal



Analyzing store transactions to identify revenue drivers, purchasing behavior, and performance metrics across branches

Dataset Scope



Transactional sales records covering multiple locations, customer categories, product types, and reward systems

Core Methodology

Executed analytical queries using Microsoft SQL Server to extract key performance indicators (KPIs) and business insights

Dataset Structure

Key Variables & Data Types

TRANSACTION & BRANCH INFO

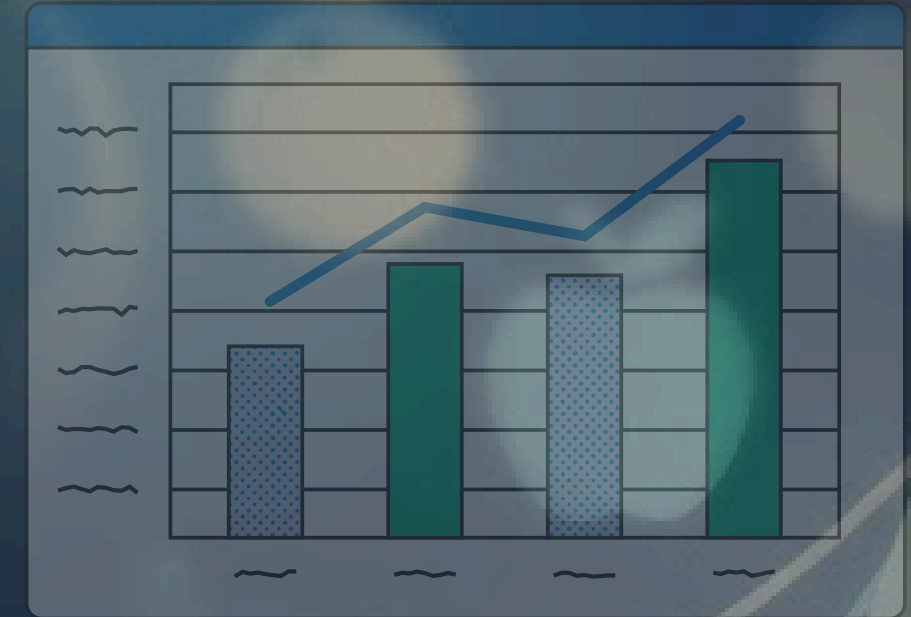
- SALE_ID: UNIQUE IDENTIFIER FOR EACH TRANSACTION.
- BRANCH & CITY: STORE LOCATIONS (BRANCH A, B, C / CITIES).
- REWARD_POINTS: LOYALTY POINTS EARNED PER TRANSACTION.

CUSTOMER & PRODUCT PROFILE

- CUSTOMER_TYPE & GENDER: DEMOGRAPHICS (MEMBER / NORMAL).
- PRODUCT_NAME: SPECIFIC ITEM PURCHASED.
- PRODUCT_CATEGORY: CLASSIFICATION OF SOLD ITEMS.

FINANCIAL & METRICS

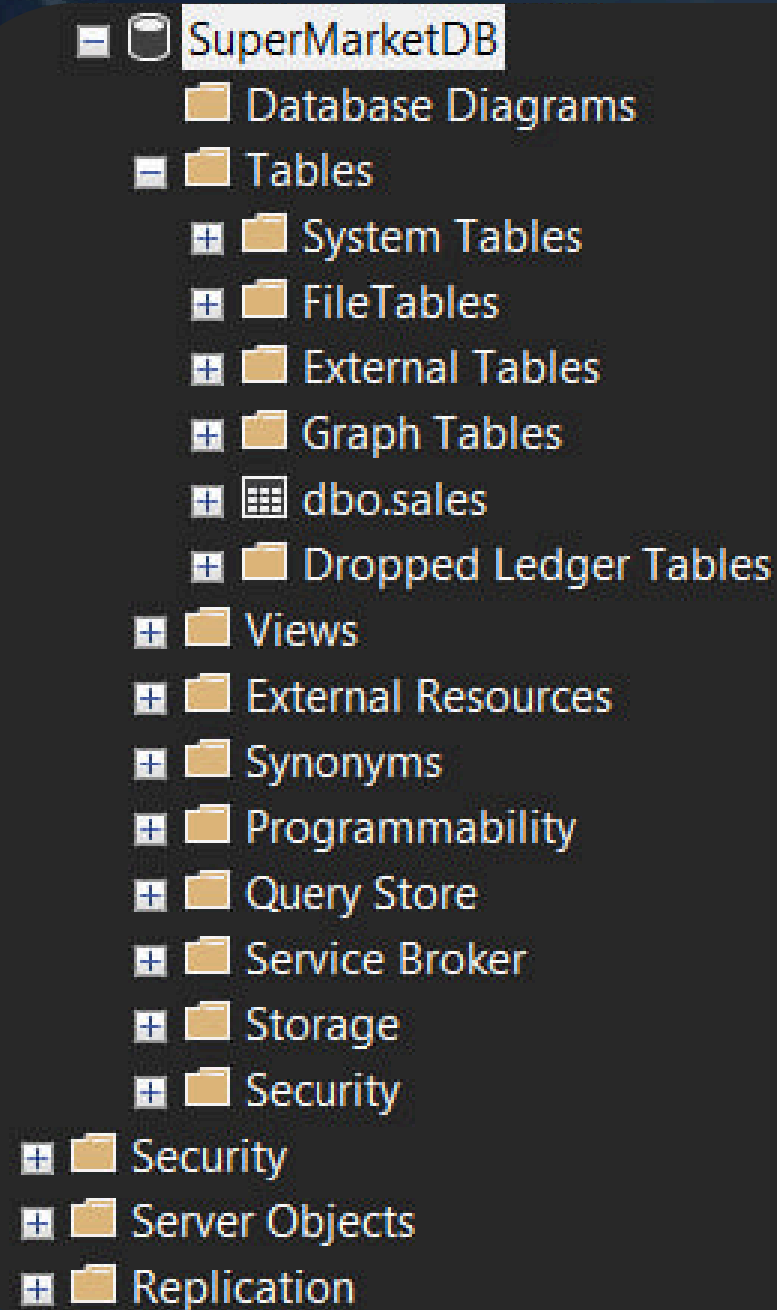
- UNIT_PRICE & QUANTITY: ITEM COST AND VOLUME PURCHASED.
- TAX: 5% TAX CALCULATED PER TRANSACTION.
- TOTAL_PRICE: FINAL AMOUNT PAID PER TRANSACTION.



Key Performance Indicators

Metric Name	Business Focus	Calculated Value	Status
Total Revenue	Overall Gross Sales	\$322.0K	High Revenue
Total Tax Collected	5% Tax Compliance	\$16.1K	Verified
Total Transactions	Customer Purchase Volume	1,000 Orders	Active
Avg Order Value	Mean Basket Size	\$322.00	Optimal

Database Setup & Environment



- Database Initialization Created SuperMarketDB on local SQL Server instance to handle relational data.
- Table Architecture Structured dbo.sales table to store sales and customer metrics.
- Environment Readiness Verified schema deployment and data integrity for multi-dimensional analysis

DATA INITIALIZATION & SCHEMA VALIDATION

RETRIEVE THE FIRST 10 ROWS FROM THE SALES TABLE TO INSPECT THE DATASET STRUCTURE.

EXTRACTED THE INITIAL 10 RECORDS FROM DBO.SALES TO VERIFY TABLE ARCHITECTURE AND SCHEMA ALIGNMENT, ENSURING PROPER DATA TYPES ACROSS ALL 12 TRANSACTIONAL ATTRIBUTES INCLUDING PRICING, TAXES, AND CUSTOMER LOYALTY METRICS.

```
2  
3 SELECT TOP 10 * FROM sales
```

100 %  No issues found

 Results  Messages

	sale_id	branch	city	customer_type	gender	product_name	product_category	unit_price	quantity	tax	total_price	reward_poin
1	1	A	New York	Member	Male	Shampoo	Personal Care	5.5	3	1.1599999666214	17.6599998474121	1
2	2	B	Los Angeles	Normal	Female	Notebook	Stationery	2.75	10	1.92999994754791	29.4300003051758	0
3	3	A	New York	Member	Female	Apple	Fruits	1.20000004768372	15	1.25999999046326	19.2600002288818	1
4	4	A	Chicago	Normal	Male	Detergent	Household	7.80000019073486	5	2.73000001907349	41.7299995422363	0
5	5	B	Los Angeles	Member	Female	Orange Juice	Beverages	3.5	7	1.72000002861023	26.2199993133545	2
6	6	A	Chicago	Normal	Male	Shampoo	Stationery	11.2399997711182	9	7.07999992370605	108.23999786377	0
7	7	A	Chicago	Normal	Male	Shampoo	Personal Care	10.710000038147	1	0.75	11.460000038147	0
8	8	B	Los Angeles	Normal	Female	Shampoo	Household	18.2299995422363	9	11.4799995422363	175.550003051758	0
9	9	A	Chicago	Member	Female	Apple	Fruits	14.1499996185303	20	19.8099994659424	302.809997558594	30
10	10	B	Los Angeles	Member	Male	Shampoo	Fruits	18.4200000762939	19	24.5	374.480010986328	37

OVERALL STORE SALES & TAX PERFORMANCE

```
13
14 SELECT
15     ROUND(SUM(total_price), 2) AS Total_Sales,
16     ROUND(SUM(tax), 2) AS Total_Tax,
17     COUNT(sale_id) AS Total_Transactions
18 FROM sales;
```

100 %  No issues found

 Results  Messages

	Total_Sales	Total_Tax	Total_Transactions
1	118583.9	7758.01	1000

CALCULATE TOTAL SALES REVENUE,
OVERALL TAX COLLECTED, AND THE TOTAL
NUMBER OF TRANSACTIONS

GENERATED \$118,583.90 IN
GROSS REVENUE ACROSS 1,000
TRANSACTIONS, WITH TOTAL
TAX COLLECTED AMOUNTING TO
\$7,758.01, REFLECTING A STABLE
BASELINE AVERAGE ORDER
VALUE OF ~\$118.58.

REGIONAL SALES PERFORMANCE BY BRANCH & CITY

CALCULATE THE TOTAL SALES FOR EACH
BRANCH AND CITY, ORDERED BY SALES
DESCENDING

```
11  
12 SELECT  
13     branch,  
14     city,  
15     ROUND(SUM(total_price), 2) AS Total_Sales  
16 FROM sales  
17 GROUP BY branch, city  
18 ORDER BY Total_Sales DESC;
```

00 %  No issues found

Results		Messages
branch	city	Total_Sales
A	Chicago	42584.71
A	New York	40226.93
B	Los Angeles	35772.26

BRANCH A IN CHICAGO LEADS
REVENUE GENERATED AT
\$42,584.71, CLOSELY FOLLOWED BY
BRANCH A IN NEW YORK AT
\$40,226.93, WHILE BRANCH B IN LOS
ANGELES GENERATED \$35,772.26.

PRODUCT CATEGORY REVENUE & ORDER ANALYSIS

```
11
12 SELECT
13     product_category,
14     ROUND(SUM(total_price), 2) AS Total_Revenue,
15     COUNT(sale_id) AS Total_Orders
16 FROM sales
17 GROUP BY product_category
18 ORDER BY Total_Revenue DESC;
```

100 % No issues found

Results Messages

	product_category	Total_Revenue	Total_Orders
1	Personal Care	27050.18	208
2	Fruits	26197.45	209
3	Beverages	22983.32	187
4	Household	21615.84	198
5	Stationery	20737.11	198

FIND TOTAL REVENUE AND TOTAL ORDERS
PER PRODUCT CATEGORY, ORDERED BY
TOTAL REVENUE DESCENDING

PERSONAL CARE LEADS ALL
CATEGORIES WITH \$27,050.18 ACROSS
208 ORDERS, CLOSELY FOLLOWED BY
FRUITS AT \$26,197.45 (209 ORDERS),
WHILE STATIONERY RANKS LOWEST IN
TOTAL REVENUE AT \$20,737.11

SALES PERFORMANCE BY CUSTOMER TYPE

```
12  
13 SELECT  
14     customer_type,  
15     COUNT(sale_id) AS Total_Orders,  
16     ROUND(SUM(total_price), 2) AS Total_Sales  
17 FROM sales  
18 GROUP BY customer_type;
```

100 %  No issues found

Results  Messages			
customer_type	Total_Orders	Total_Sales	
Normal	484	55370.27	
Member	516	63213.63	

CALCULATE THE TOTAL ORDERS AND
TOTAL SALES GENERATED BY EACH
CUSTOMER TYPE

MEMBERS GENERATED HIGHER TOTAL
REVENUE AT \$63,213.63 ACROSS 516
ORDERS, OUTPERFORMING NORMAL
CUSTOMERS WHO GENERATED
\$55,370.27 OVER 484 ORDER

REVENUE & ORDER ANALYSIS BY GENDER

FIND TOTAL ORDERS AND TOTAL SALES
GENERATED ACROSS DIFFERENT GENDER
CATEGORIES

```
12  
13 SELECT  
14     gender,  
15     COUNT(sale_id) AS Total_Orders,  
16     ROUND(SUM(total_price), 2) AS Total_Sales  
17 FROM sales  
18 GROUP BY gender;
```

% ☐ No issues found

Results			Messages
gender	Total_Orders	Total_Sales	
Male	528	64318.45	
Female	472	54265.45	

MALE CUSTOMERS GENERATED
\$64,318.45 IN REVENUE ACROSS 528
ORDERS, OUTPERFORMING FEMALE
CUSTOMERS WHO GENERATED
\$54,265.45 OVER 472 ORDERS

AVERAGE UNIT PRICE BY CATEGORY

```
13
14 SELECT
15     product_category,
16     ROUND(AVG(unit_price), 2) AS Avg_Unit_Price
17 FROM sales
18 GROUP BY product_category
19 ORDER BY Avg_Unit_Price DESC;
```

00 %  No issues found

Results Messages

	product_category	Avg_Unit_Price
1	Personal Care	11.25
2	Beverages	11.11
3	Household	10.93
4	Fruits	10.85
5	Stationery	10.04

CALCULATE THE AVERAGE UNIT PRICE FOR EACH PRODUCT CATEGORY, ORDERED DESCENDING

PERSONAL CARE HOLDS THE HIGHEST AVERAGE UNIT PRICE AT \$11.25, FOLLOWED BY BEVERAGES AT \$11.11, WHILE STATIONERY RECORDS THE LOWEST AVERAGE PRICING ACROSS CATEGORIES AT \$10.04.

TOP SELLING PRODUCTS BY QUANTITY & REVENUE

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```
SELECT
    product_name,
    SUM(quantity) AS Total_Quantity_Sold,
    ROUND(SUM(total_price), 2) AS Total_Sales
FROM sales
GROUP BY product_name
ORDER BY Total_Quantity_Sold DESC;
```

0 %

✓ No issues found

ResultsMessages

product_name	Total_Quantity_Sold	Total_Sales
Shampoo	2238	27041.36
Orange Juice	2183	24686.46
Notebook	2165	24792.98
Detergent	2010	22449.07
Apple	1741	19614.03

DETERMINE TOTAL QUANTITY SOLD AND TOTAL REVENUE GENERATED FOR EACH PRODUCT, ORDERED BY TOTAL UNITS SOLD DESCENDING.

SHAMPOO RANKS AS THE TOP-PERFORMING PRODUCT WITH 2,238 UNITS SOLD GENERATING \$27,041.36 IN SALES, FOLLOWED BY ORANGE JUICE (2,183 UNITS / \$24,686.46), WHILE APPLE RECORDS THE LOWEST SALES VOLUME AT 1,741 UNITS (\$19,614.03)

HIGHEST VALUE INDIVIDUAL TRANSACTIONS

```
15
16 SELECT TOP 5
17     sale_id, branch, city, product_name, total_price
18 FROM sales
19 ORDER BY total_price DESC;
```

0 % No issues found

	sale_id	branch	city	product_name	total_price
	958	A	New York	Notebook	433.989990234375
	58	A	Chicago	Notebook	427.140014648438
	161	A	Chicago	Orange Juice	425.100006103516
	622	A	New York	Detergent	423.079986572266
	761	B	Los Angeles	Apple	421.640014648438

RETRIEVE THE TOP 5 HIGHEST-VALUE TRANSACTIONS, SHOWING SALE ID, BRANCH, CITY, PRODUCT NAME, AND TOTAL PRICE.

TRANSACTION ID 958 IN NEW YORK RECORDED THE HIGHEST SINGLE PURCHASE VALUE AT \$433.99 FOR NOTEBOOKS, WITH 4 OUT OF THE TOP 5 TRANSACTIONS ORIGINATING FROM BRANCH A ACROSS NEW YORK AND CHICAGO.

REWARD POINTS ACCUMULATION BY CUSTOMER TYPE

CALCULATE THE TOTAL REWARD POINTS ACCUMULATED BY EACH CUSTOMER TYPE.

MEMBERS ACCUMULATED 100% OF LOYALTY POINTS (6,057 TOTAL POINTS), WHEREAS NORMAL NON-MEMBER CUSTOMERS HOLD 0 POINTS, VALIDATING THE PROGRAM'S STRICT RESTRICTION TO REGISTERED MEMBER ACCOUNTS.

```
14
15 SELECT
16     customer_type,
17     SUM(reward_points) AS Total_Reward_Points
18 FROM sales
19 GROUP BY customer_type;
```

00 % No issues found

Results Messages

customer_type	Total_Reward_Points
Normal	0
Member	6057

PRODUCT SALES BREAKDOWN IN NEW YORK

12

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```
SELECT
    product_name,
    ROUND(SUM(total_price), 2) AS NYC_Sales
FROM sales
WHERE city = 'New York'
GROUP BY product_name
ORDER BY NYC_Sales DESC;
```

100 %

✓ No issues found

Results

Messages

	product_name	NYC_Sales
1	Shampoo	9879.68
2	Notebook	9251.19
3	Detergent	8120.24
4	Apple	6977.13
5	Orange Juice	5998.69

CALCULATE THE TOTAL SALES GENERATED BY EACH PRODUCT SPECIFICALLY IN NEW YORK, ORDERED DESCENDING

SHAMPOO LEADS SALES PERFORMANCE IN NEW YORK WITH \$9,879.68, FOLLOWED CLOSELY BY NOTEBOOKS (\$9,251.19), WHILE ORANGE JUICE RECORDS THE LOWEST REGIONAL SALES CONTRIBUTION AT \$5,998.69.

PRODUCT CATEGORIES WITH TOTAL QUANTITY EXCEEDING 10 UNITS

```
12
13 SELECT
14     product_category,
15     SUM(quantity) AS Total_Quantity
16 FROM sales
17 GROUP BY product_category
18 HAVING SUM(quantity) > 10
19 ORDER BY Total_Quantity DESC;
```

100 %  No issues found

 Results  Messages

	product_category	Total_Quantity
1	Fruits	2286
2	Personal Care	2278
3	Beverages	1952
4	Stationery	1913
5	Household	1908

FILTER AND DISPLAY PRODUCT CATEGORIES WHERE TOTAL QUANTITY SOLD EXCEEDS 10 UNITS, ORDERED BY VOLUME DESCENDING.

ALL 5 PRODUCT CATEGORIES EASILY SURPASSED THE 10-UNIT THRESHOLD, LED BY FRUITS WITH 2,286 UNITS AND PERSONAL CARE WITH 2,278 UNITS, SHOWING HIGH OVERALL VOLUME DEMAND ACROSS EVERY SEGMENT

REWARD POINTS DISTRIBUTION BY BRANCH & CITY

```
13 SELECT
14     branch,
15     city,
16     SUM(reward_points) AS Total_Branch_Points
17 FROM sales
18 GROUP BY branch, city
19 ORDER BY Total_Branch_Points DESC;
```

100 % No issues found

	branch	city	Total_Branch_Points
1	A	Chicago	2284
2	A	New York	2061
3	B	Los Angeles	1712

CALCULATE TOTAL REWARD POINTS
ACCUMULATED BY BRANCH AND CITY,
ORDERED DESCENDING.

BRANCH A IN CHICAGO LEADS LOYALTY
POINT ACCUMULATION WITH 2,284
POINTS, FOLLOWED BY BRANCH A IN NEW
YORK AT 2,061 POINTS, WHILE BRANCH B
IN LOS ANGELES ACCOUNTS FOR 1,712
POINTS

EXECUTIVE SUMMARY & STRATEGIC RECOMMENDATIONS

- DRIVE LOYALTY ENROLLMENT: CONVERT "NORMAL" CUSTOMERS INTO "MEMBERS" TO CAPTURE HIGHER ORDER VOLUME AND DRIVE REPEAT PURCHASES THROUGH THE 100% MEMBER-EXCLUSIVE REWARD PROGRAM.
- CAPITALIZE ON TOP PERFORMING REGIONS: FOCUS INVENTORY EXPANSION AND MARKETING INVESTMENTS ON BRANCH A (CHICAGO & NEW YORK), WHICH GENERATES OVER 70% OF HIGH-VALUE TRANSACTIONS.
- OPTIMIZE INVENTORY & PRICING: PRIORITIZE STOCK FOR TOP DRIVERS (SHAMPOO & ORANGE JUICE) WHILE EVALUATING PRICE ELASTICITY FOR LOWER-VOLUME CATEGORIES LIKE STATIONERY.

THANK
YOU!

